

Exercise Sheet 3 — One-Way ANOVA

Study Guide: Worked Solutions with Explanations

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How to use this guide: Each question is answered step by step, with plain-English explanations *before* the formal answer. Code is shown so you can reproduce every result in R. Leave the annotation boxes empty — those are for your own notes later!

1 Key Topics Covered

- **Experimental design vocabulary** — factors, levels, treatments
 - **Randomized block design** — what it is and when to use it
 - **Multiple testing problem** — Type I error accumulation
 - **ANOVA assumptions** — what needs to hold before you run the test
 - **One-Way ANOVA** — hypothesis test, ANOVA table, F-statistic
 - **Effect size** ² — how big is the effect, not just is it significant?
 - **Tukey’s HSD post hoc test** — which groups are actually different?
 - **ANOVA model** — writing the regression-style equation
 - **Model diagnostics** — checking your assumptions after fitting
 - **Interpreting outputs** — Fligner-Killeen, Levene, diagnostic plots
-

2 Core Concepts Explained

2.1 What is ANOVA? (The Big Picture)

Imagine you want to know whether three different fertilizers produce different crop yields. You could run lots of t-tests comparing them two at a time, but that inflates your chance of getting a false positive. **ANOVA (Analysis of Variance)** solves this — it compares *all* groups in a single test by asking: “Is the variation *between* groups larger than the variation *within* groups?”

If yes, something is going on. If no, any group differences are probably just random noise.

2.2 Experimental Design Vocabulary

Term	Plain English	Formal Definition
Factor	The thing you're changing on purpose	A controlled explanatory variable whose levels are set by the experimenter
Level	The specific settings of that thing	Different values of the controlled variable
Treatment	Each unique experimental condition	Each combination of factor levels

Memory trick: A *factor* is like a dial on a machine. The *levels* are the positions you can turn it to. A *treatment* is one specific dial position (or combination of positions if there are multiple dials).

2.3 The ANOVA Model

The ANOVA model looks like this:

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

Symbol	Meaning
y_{ij}	The observation for subject j in group i
μ	The overall mean (grand mean)
τ_i	The effect of treatment i (how much group i differs from overall mean)
ε_{ij}	Random error — the bit we can't explain

In R's regression-style output, this becomes:

$$\hat{y}_{ij} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots$$

Where $\hat{\beta}_0$ is the mean of the **reference group** (usually group 1), and the $\hat{\beta}$ values are how much each other group differs from it.

2.4 The ANOVA Table — Decoded

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Treatment	k-1	SS_T	MS_T	F	p
Residuals	N-k	SS_E	MS_E		

- **Df** = degrees of freedom. For the treatment: number of groups minus 1. For residuals: total observations minus number of groups.
- **Sum Sq** = total variability attributable to this source.
- **Mean Sq** = Sum Sq / Df (average variability per degree of freedom).
- **F value** = MS_Treatment / MS_Residuals. Big F = groups are more different than random noise.
- **Pr(>F)** = p-value. If < 0.05 , we reject H_0 (at least one group mean differs).

2.5 Effect Size ² (Eta-Squared)

The p-value tells you *if* an effect exists. ² tells you *how big* it is.

$$\eta^2 = \frac{SS_{Treatment}}{SS_{Total}} = \frac{SS_{Treatment}}{SS_{Treatment} + SS_{Residuals}}$$

² value	Interpretation
~0.01	Small effect
~0.06	Medium effect
~0.14+	Large effect

² is literally “the proportion of total variation explained by the treatment.”

2.6 Tukey’s HSD — The Pairwise Test

After ANOVA tells you *something* is different, Tukey’s Honestly Significant Difference (HSD) test tells you *which pairs* are different. It controls the family-wise error rate (the chance of any false positive across all comparisons).

Reading Tukey output: Look at p adj. If it’s < 0.05 , those two groups are significantly different.

2.7 Model Diagnostics — The Four Plots

When you run `plot(model)` in R, you get four plots:

1. **Residuals vs Fitted:** Should show random scatter with no pattern. A pattern = violated assumptions.
 2. **Q-Q Plot:** Residuals should fall on the diagonal line. Deviations = non-normality.
 3. **Scale-Location:** Should be flat. A funnel shape = unequal variances (heteroscedasticity).
 4. **Residuals vs Leverage:** Points with high leverage AND high residuals are *influential* — they may be pulling your results.
-

3 Questions and Worked Solutions

3.1 Q1 — Experimental Design Vocabulary

3.1.1 Question

In the context of experimental design, what is: (i) a factor? (ii) a level? (iii) a treatment?

3.1.2 Step-by-step explanation

Think of making coffee. You decide to test whether grind size and water temperature affect taste.

- The things you're **controlling** (grind size, water temperature) → **factors**
- The specific settings you try (coarse/fine/medium, 85°C/90°C/95°C) → **levels**
- Each specific combination you actually brew (fine + 90°C, coarse + 95°C, etc.) → **treatment**

3.1.3 Answer

(i) **Factor:** A factor is a controlled explanatory variable — a variable whose levels are deliberately set by the experimenter.

(ii) **Level:** A level represents a specific value (or category) of a factor.

(iii) **Treatment:** A treatment is one specific experimental condition. If there is only one factor, each level is a treatment. If there are multiple factors, each **combination** of factor levels is a treatment.

Annotation space — add your own notes here:

3.2 Q2 — Drill Press Experiment

3.2.1 Question

A mechanical engineer studies thrust force developed by a drill press. He suspects drilling speed and feed rate affect thrust force. He selects 4 feed rates and 2 drilling speeds (high and low).

(i) How many factors? (ii) How many levels each? (iii) How many treatments?

3.2.2 Step-by-step explanation

Count what is being deliberately varied:

- **Factor 1:** Drilling speed (2 levels: high, low)
- **Factor 2:** Feed rate (4 levels)
- **Treatments:** Every possible combination = $2 \times 4 = 8$ treatments

3.2.3 Answer

- (i) Two factors: drilling speed and feed rate.
(ii) Drilling speed has 2 levels (high, low). Feed rate has 4 levels.
(iii) $2 \times 4 = 8$ treatments total.

Annotation space:

3.3 Q3 & Q4 — Randomized Block Design

3.3.1 Question

Give an example of a randomized block design. Why is it appropriate?

3.3.2 Plain English Explanation

A **randomized block design** is used when you know there's a nuisance variable that could affect your results, but you're not interested in it — you just want to *control* for it.

Example: A manufacturer wants to test 4 chemical formulations. Each formulation is made from a batch of raw material, but each batch is only large enough for 4 formulations. Different batches might have natural variation. So “batch” is a nuisance variable.

Design:

Formula	Batch 1	Batch 2	Batch 3	Batch 4	Batch 5
1					
2					
3					
4					

Each batch = one “block.” Within each block, formulas are randomly assigned. This controls for batch-to-batch variation.

3.3.3 Answer

Q3: A manufacturer testing 4 formulations across 5 batches of raw material, where each batch introduces natural variation, uses a randomized block design (see table above).

Q4: The randomized block design is appropriate because it **controls for the effect of batch variation**, which would otherwise mask or confound the treatment effects.

Annotation space:

3.4 Q5 — Multiple Testing and Type I Error

3.4.1 Question

If we test **five** null hypotheses (all true) at significance level $\alpha = 0.01$, what is the probability of making **at least one** Type I error?

3.4.2 Plain English Explanation

A **Type I error** is a false positive — rejecting H_0 when it's actually true. At $\alpha = 0.01$, each individual test has a 1% chance of a false positive.

The trick is to use the complement: instead of calculating $P(\text{at least one error})$, calculate $P(\text{no errors at all})$ and subtract from 1.

$$P(\text{at least one error}) = 1 - P(\text{no errors})$$

$$P(\text{no error on one test}) = 1 - 0.01 = 0.99$$

$$P(\text{no errors on 5 tests}) = 0.99^5 = 0.9510$$

$$P(\text{at least one error}) = 1 - 0.9510 = \mathbf{0.049}$$

3.4.3 Answer

$$P(\text{at least one Type I error}) = 1 - (1 - 0.01)^5 = 1 - 0.99^5 \approx \mathbf{0.049}$$

Even though each test had only a 1% false positive rate, running 5 tests brings the combined risk up to nearly 5%! This is why we need methods like Tukey's HSD when making multiple comparisons.

Annotation space:

3.5 Q6 — ANOVA Assumptions

3.5.1 Question

Name two assumptions that should be satisfied before carrying out an ANOVA.

3.5.2 Answer

1. **Independence (Random Sampling):** Observations must be independent of each other. No observation should influence another.
2. **Homogeneity of Variance (Equal Variances):** The variance within each group should be approximately equal across all groups. Tested using the **Fligner-Killeen** or **Levene** test.

Bonus (not always tested but important to know): Normality — the residuals (errors) should be approximately normally distributed. This is checked via the Q-Q plot in model diagnostics.

Annotation space:

3.6 Q7 — Kenton Foods: Full ANOVA Worked Example

3.6.1 Question

Kenton Food Company tests 4 package designs for cereal across 20 stores (5 stores per design). Analyse the R output provided.

3.6.2 Setup and Data

```
library(tidyverse)

# Load the data
kenton <- read_tsv("Kenton_Foods.txt")
kenton$Package <- as.factor(kenton$Package)

# Quick look
glimpse(kenton)
```

Rows: 20

Columns: 2

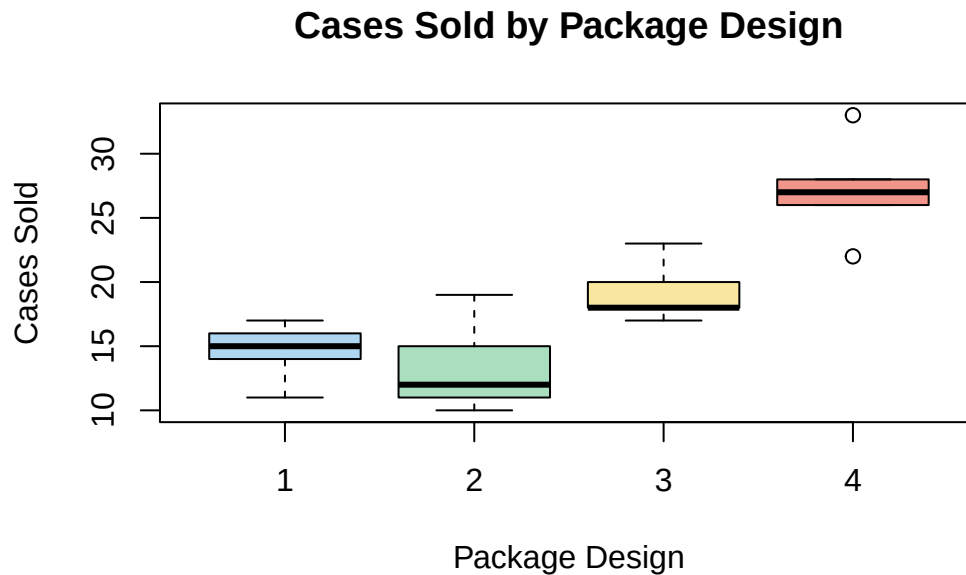
```
$ Cases    <dbl> 11, 17, 16, 14, 15, 12, 10, 15, 19, 11, 23, 20, 18, 17, 18, 27~
$ Package  <fct> 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4
```

3.6.3 Step 1: Exploratory Data Analysis

```
# Summary statistics by package
kenton %>%
  group_by(Package) %>%
  summarise(
    mean_Cases = mean(Cases),
    sd_Cases   = sd(Cases),
    n          = n()
  )
```

```
# A tibble: 4 x 4
  Package mean_Cases sd_Cases    n
  <fct>      <dbl>    <dbl> <int>
1 1         14.6      2.30     5
2 2         13.4      3.65     5
3 3         19.2      2.39     5
4 4         27.2      3.96     5
```

```
# Boxplot
boxplot(kenton$Cases ~ kenton$Package,
        xlab = "Package Design",
        ylab = "Cases Sold",
        main = "Cases Sold by Package Design",
        col  = c("#AED6F1", "#A9DFBF", "#F9E79F", "#F1948A"))
```



How to read the boxplot: Each box shows the middle 50% of sales for that package. The line in the middle is the median. Points outside the “whiskers” are potential outliers. Here, Package 4 clearly has higher sales.

3.6.4 Step 2: Q7(i) — Fligner-Killeen Test

3.6.4.1 Plain English

Before running ANOVA, we must check that the **variance in sales is similar** across all package designs. The Fligner-Killeen test does this.

- H_0 : all groups have equal variance
- H_a : at least one group has different variance

If $p > 0.05$, we’re fine to proceed.

```
fligner.test(kenton$Cases ~ kenton$Package)
```

Fligner-Killeen test of homogeneity of variances

data: kenton\$Cases by kenton\$Package

Fligner-Killeen:med chi-squared = 1.1321, df = 3, p-value = 0.7693

Result from output: Fligner-Killeen: med chi-squared = 1.1321, df = 3, **p-value = 0.7693**

Interpretation: $p = 0.7693 \gg 0.05$. We **fail to reject H_0** . There is no significant difference in variance across package designs. The equal variance assumption is satisfied — we can proceed with ANOVA.

3.6.5 Step 3: Q7(ii) — Hypotheses

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

“The mean number of cases sold is the same for all four package designs.”

$$H_A : \text{At least one } \mu_i \text{ is different}$$

“At least one package design leads to a different mean number of cases sold.”

3.6.6 Step 4: Q7(iii) — Degrees of Freedom

Why does Package have 3 degrees of freedom?

The formula is: $df = (\text{number of groups}) - 1 = 4 - 1 = \mathbf{3}$

Think of it this way: once you know 3 group means and the overall mean, the 4th group mean is already determined. You “use up” one df by estimating the overall mean.

3.6.7 Step 5: Q7(iv) — Run the ANOVA

```
model_kenton <- aov(kenton$Cases ~ kenton$Package)
summary(model_kenton)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
kenton\$Package	3	586.8	195.6	19.56	1.33e-05 ***
Residuals	16	160.0	10.0		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Reading the ANOVA Table:

Source	Df	Sum Sq	Mean Sq	F value	p-value
Package	3	586.8	195.6	19.56	1.33e-05 ***
Residuals	16	160.0	10.0		

- **F(3, 16) = 19.56** — The between-group variation is nearly 20× the within-group noise. That's huge.
- **p < 0.001** — Extremely strong evidence against H₀.

Effect Size:

$$\eta^2 = \frac{SS_{Package}}{SS_{Package} + SS_{Residuals}} = \frac{586.8}{586.8 + 160.0} = \frac{586.8}{746.8} \approx \mathbf{0.786}$$

This means **78.6% of the total variation** in cases sold is explained by the package design. That is a very large effect.

3.6.8 Step 6: Tukey's HSD — Pairwise Comparisons

```
TukeyHSD(model_kenton)
```

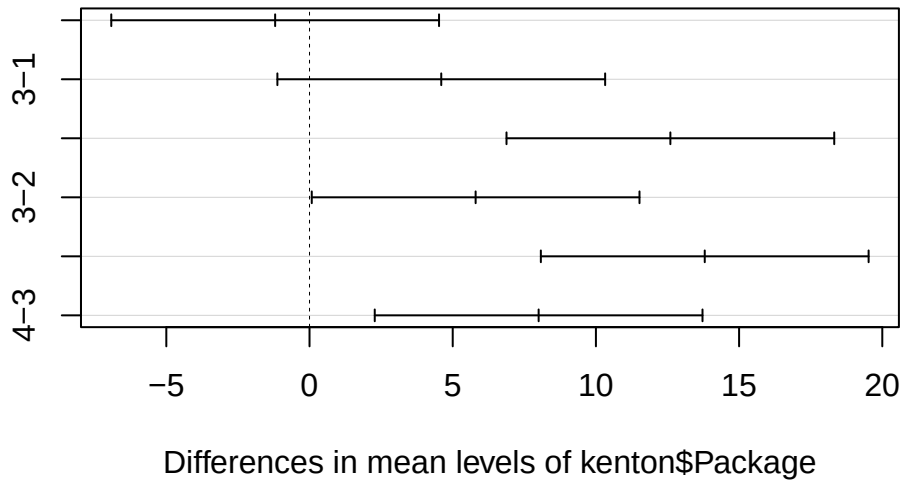
```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = kenton$Cases ~ kenton$Package)
```

```
$`kenton$Package`
      diff      lwr      upr      p adj
2-1 -1.2 -6.92203965  4.52204 0.9305700
3-1  4.6 -1.12203965 10.32204 0.1395792
4-1 12.6  6.87796035 18.32204 0.0000569
3-2  5.8  0.07796035 11.52204 0.0463912
4-2 13.8  8.07796035 19.52204 0.0000194
4-3  8.0  2.27796035 13.72204 0.0051230
```

```
plot(TukeyHSD(model_kenton))
```

95% family-wise confidence level



Reading the Tukey output:

Comparison	Difference	95% CI Lower	95% CI Upper	p adj	Significant?
2 vs 1	-1.2	-6.92	4.52	0.931	No
3 vs 1	4.6	-1.12	10.32	0.140	No
4 vs 1	12.6	6.88	18.32	< 0.001	Yes
3 vs 2	5.8	0.08	11.52	0.046	Yes
4 vs 2	13.8	8.08	19.52	< 0.001	Yes
4 vs 3	8.0	2.28	13.72	0.005	Yes

How to interpret the Tukey plot: A confidence interval that **does not cross zero** = significant difference. If the CI includes zero, the two groups could plausibly have the same mean.

Summary grouping table (for exam reports):

Package	Mean Cases	Grouping
1	14.6	A
2	13.4	A
3	19.2	B
4	27.2	C

Groups sharing a letter are NOT significantly different from each other.

Full written report (exam-style):

A one-way analysis of variance indicated that package design has a significant effect on the number of cases of cereal sold, $F(3, 16) = 19.56$, $MSE = 10.0$, $p < 0.001$, $\eta^2 = 0.786$. Tukey's HSD test indicated that the mean number of cases sold was significantly different between all designs **except** designs 1 vs 2 ($p = 0.931$) and designs 1 vs 3 ($p = 0.140$). Package 4 had the highest mean sales ($M = 27.2$).

3.6.9 Step 7: Q7(v) — The ANOVA Model

```
summary.lm(model_kenton)
```

Call:

```
aov(formula = kenton$Cases ~ kenton$Package)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.20	-1.60	-0.40	1.45	5.80

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	14.600	1.414	10.32	1.76e-08	***
kenton\$Package2	-1.200	2.000	-0.60	0.5569	
kenton\$Package3	4.600	2.000	2.30	0.0352	*
kenton\$Package4	12.600	2.000	6.30	1.06e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.162 on 16 degrees of freedom

Multiple R-squared: 0.7858, Adjusted R-squared: 0.7456

F-statistic: 19.56 on 3 and 16 DF, p-value: 1.333e-05

From the regression-style output:

$$\hat{y}_{ij} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3$$

$$\hat{y}_{ij} = 14.6 - 1.2x_1 + 4.6x_2 + 12.6x_3$$

What each term means:

Term	Value	Meaning
$\hat{\beta}_0 = 14.6$	Intercept	Mean cases sold for Package 1 (the reference group)
$\hat{\beta}_1 = -1.2$	Package 2 effect	Package 2 sells 1.2 fewer cases on average than Package 1
$\hat{\beta}_2 = 4.6$	Package 3 effect	Package 3 sells 4.6 more cases on average than Package 1
$\hat{\beta}_3 = 12.6$	Package 4 effect	Package 4 sells 12.6 more cases on average than Package 1

The dummy variable coding:

Group	x_1	x_2	x_3
Package 1	0	0	0
Package 2	1	0	0
Package 3	0	1	0
Package 4	0	0	1

Annotation space:

3.7 Q8 — Vitamin C in Multivitamins

3.7.1 Question

Twelve measurements of Vitamin C content (mg) were taken for each of three multivitamin brands (A, B, C). Interpret the R output.

3.7.2 Plain English Setup

We're asking: "Do these three vitamin brands actually deliver different amounts of Vitamin C, or is any difference just random variation?"

3.7.3 Q8(i) — Hypotheses

$$H_0 : \mu_A = \mu_B = \mu_C$$

“The mean Vitamin C content is the same across all three brands.”

$$H_A : \text{At least one brand has a different mean Vitamin C content}$$

3.7.4 Q8(ii) — Levene Test Hypotheses

$$H_0 : \sigma_A^2 = \sigma_B^2 = \sigma_C^2$$

“The variance in Vitamin C content is equal across all three brands.”

$$H_A : \text{At least one brand has different variance}$$

3.7.5 Q8(iii) — Interpreting the Levene Test

From the output:

```
Levene's Test: Df F value Pr(>F)
group    2    0.6815    0.5128
      33
```

Interpretation: $p = 0.5128 \gg 0.05$. We **fail to reject H_0** . There is not enough evidence to suggest that variances differ across brands. The homogeneity of variance assumption is met — we can proceed with ANOVA.

3.7.6 Q8(iv) — Completing the ANOVA Table

The original output had missing values:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
variety	-	29.8	-	-	0.355
Residuals	-	459.4	-		

Filling in the blanks:

- **variety Df:** 3 groups $- 1 = 2$

- **Residuals Df:** 36 total – 3 groups = **33**
- **variety Mean Sq:** 29.8 / 2 = **14.9**
- **Residuals Mean Sq:** 459.4 / 33 = **13.92**
- **F value:** 14.9 / 13.92 = **1.069** (matches the given value)

Completed ANOVA table:

Source	Df	Sum Sq	Mean Sq	F value	p-value
variety	2	29.8	14.9	1.069	0.355
Residuals	33	459.4	13.92		

Effect Size:

$$\eta^2 = \frac{29.8}{29.8 + 459.4} = \frac{29.8}{489.2} \approx \mathbf{0.061}$$

A small effect (6% of variation explained by brand).

Written report:

A one-way analysis of variance did not provide evidence to suggest that the mean Vitamin C content varied between multivitamin varieties, $F(2, 33) = 1.069$, $MSE = 13.92$, $p = 0.355$, $\eta^2 = 0.061$. Tukey's HSD test confirmed that no pairwise comparisons were significantly different at the 5% level.

3.7.7 Q8(v) — ANOVA Model

$$\hat{y}_{ij} = 147.254 + 1.949x_1 + 1.907x_2 + e_{ij}$$

Term	Value	Meaning
147.254	Intercept	Mean Vitamin C for Variety A (reference group)
1.949	Variety B effect	Variety B has 1.949 mg more Vitamin C than A (not significant)
1.907	Variety C effect	Variety C has 1.907 mg more Vitamin C than A (not significant)
e_{ij}	Residual	The unexplained random variation

Annotation space:

3.8 Q9 — Soft Drink End-Aisle Displays

3.8.1 Question

Three end-aisle display designs were tested in 15 stores (5 per design). The response is percentage increase in sales. Perform a full analysis.

3.8.2 Setup

```
drinks <- read_tsv("soft_drink.txt")
drinks$Display <- as.factor(drinks$Display)
glimpse(drinks)
```

Rows: 15

Columns: 2

\$ Increase <dbl> 5.43, 5.71, 6.22, 6.01, 5.29, 6.24, 6.71, 5.98, 5.66, 6.60, 8~

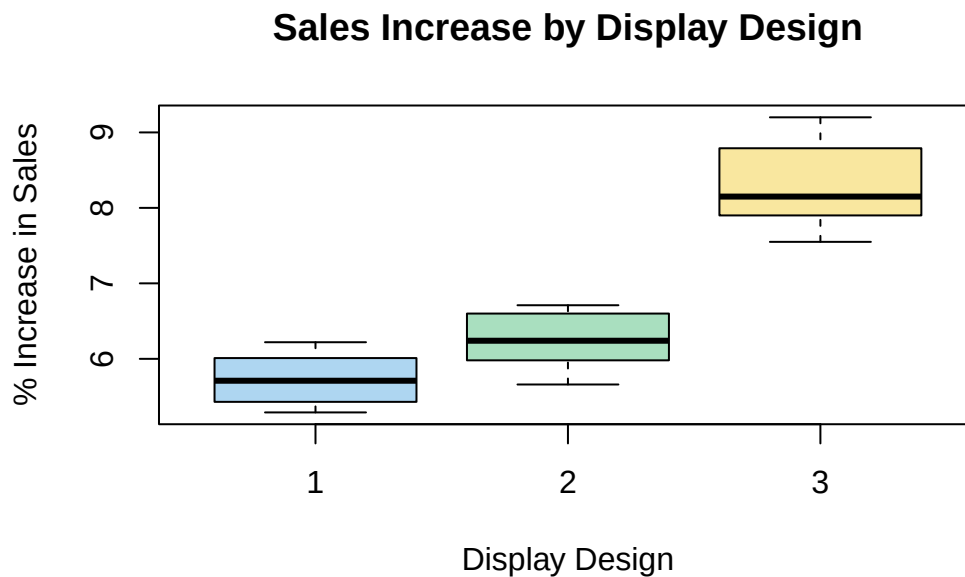
\$ Display <fct> 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3

3.8.3 Q9(i) — Exploratory Data Analysis

```
# Summary statistics
drinks %>%
  group_by(Display) %>%
  summarise(
    mean_Increase = mean(Increase),
    sd_Increase   = sd(Increase),
    max_Increase  = max(Increase),
    min_Increase  = min(Increase),
    n             = n()
  )
```

```
# A tibble: 3 x 6
  Display mean_Increase sd_Increase max_Increase min_Increase     n
  <fct>      <dbl>      <dbl>      <dbl>      <dbl> <int>
1 1          5.73        0.388        6.22        5.29     5
2 2          6.24        0.434        6.71        5.66     5
3 3          8.32        0.670        9.2         7.55     5
```

```
# Boxplot
boxplot(drinks$Increase ~ drinks$Display,
        xlab = "Display Design",
        ylab = "% Increase in Sales",
        main = "Sales Increase by Display Design",
        col = c("#AED6F1", "#A9DFBF", "#F9E79F"))
```



Summary statistics table:

Design	Mean % Increase	SD	Max	Min	n
1	5.73	0.388	6.22	5.29	5
2	6.24	0.434	6.71	5.66	5
3	8.32	0.670	9.20	7.55	5

How to interpret the boxplot: Display 3 clearly has higher median sales. Displays 1 and 2 look similar. No obvious outliers. This already suggests Display 3 might be the winner.

Check for equal variances:

```
fligner.test(drinks$Increase ~ drinks$Display)
```

Fligner-Killeen test of homogeneity of variances

data: drinks\$Increase by drinks\$Display

Fligner-Killeen: med chi-squared = 2.0427, df = 2, p-value = 0.3601

Fligner-Killeen: p = 0.3601. Since $p > 0.05$, variances are equal across groups.

3.8.4 Q9(ii) — ANOVA Table

```
model_drinks <- aov(drinks$Increase ~ drinks$Display)
summary(model_drinks)
```

```
              Df Sum Sq Mean Sq F value    Pr(>F)
drinks$Display  2  18.78   9.392   35.77 8.78e-06 ***
Residuals     12   3.15   0.263
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA Table:

Source	Df	Sum Sq	Mean Sq	F value	p-value
Display	2	18.78	9.392	35.77	< 0.001
Residuals	12	3.15	0.263		

Interpreting this table: $F(2, 12) = 35.77$. The F-statistic is very large — the between-group variation is $35\times$ the within-group noise. $p < 0.001 \rightarrow$ very strong evidence that at least one display performs differently.

Effect Size:

$$\eta^2 = \frac{18.78}{18.78 + 3.15} = \frac{18.78}{21.93} \approx \mathbf{0.856}$$

Massive effect — 85.6% of variation in sales is explained by which display was used.

3.8.5 Q9(iii) — Pairwise Comparisons (Tukey HSD)

```
TukeyHSD(model_drinks)
```

```
Tukey multiple comparisons of means  
95% family-wise confidence level
```

```
Fit: aov(formula = drinks$Increase ~ drinks$Display)
```

```
$`drinks$Display`  
      diff      lwr      upr    p adj  
2-1 0.506 -0.358547 1.370547 0.2988893  
3-1 2.586  1.721453 3.450547 0.0000107  
3-2 2.080  1.215453 2.944547 0.0000909
```

```
plot(TukeyHSD(model_drinks))
```



Tukey results:

Comparison	Difference	95% CI	p adj	Significant?
2 vs 1	0.506	(−0.47, 1.48)	0.358	No
3 vs 1	2.586	(1.61, 3.56)	< 0.001	Yes
3 vs 2	2.080	(1.11, 3.05)	0.001	Yes

Summary grouping table:

Design	Mean % Increase	SD	Group
1	5.73	0.388	A
2	6.24	0.434	A
3	8.32	0.670	B

Designs 1 and 2 are not significantly different from each other (same letter A), but both are significantly different from Design 3.

3.8.6 Q9(v) — ANOVA Model

```
summary.lm(model_drinks)
```

Call:

```
aov(formula = drinks$Increase ~ drinks$Display)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-0.768 -0.360 -0.022  0.417  0.882
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    5.7320     0.2291  25.015 1.01e-11 ***
drinks$Display2  0.5060     0.3241   1.561   0.144
drinks$Display3  2.5860     0.3241   7.980 3.86e-06 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 0.5124 on 12 degrees of freedom

Multiple R-squared: 0.8564, Adjusted R-squared: 0.8324

F-statistic: 35.77 on 2 and 12 DF, p-value: 8.782e-06

$$\hat{y}_{ij} = 5.732 + 0.506x_1 + 2.586x_2$$

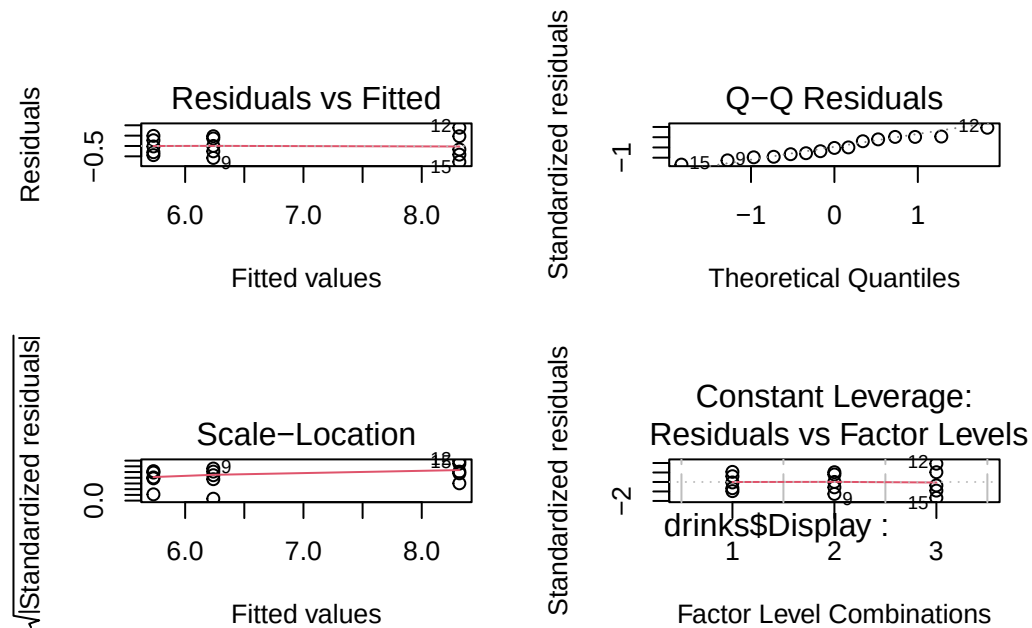
Term	Meaning
5.732	Mean % increase for Display 1 (reference)
0.506	Display 2 sells 0.506% more than Display 1 (not significant)
2.586	Display 3 sells 2.586% more than Display 1 (significant)

Dummy variable coding:

Group	x_1	x_2
Display 1	0	0
Display 2	1	0
Display 3	0	1

3.8.7 Q9(vi) — Model Diagnostics

```
par(mfrow = c(2, 2))
plot(model_drinks)
```



How to read each diagnostic plot:

- 1. Residuals vs Fitted:** The points should be scattered randomly around the horizontal line at zero, with no obvious pattern. A U-shape or funnel would indicate a problem. Here the scatter looks random →
- 2. Normal Q-Q Plot:** The points should follow the diagonal dashed line closely. If they curve away (especially at the ends), the normality assumption is violated. Here the points sit close to the line →
- 3. Scale-Location Plot:** The red line should be roughly flat. An upward slope means higher fitted values have more variance (heteroscedasticity). Here it looks approximately flat →
- 4. Residuals vs Leverage:** High-leverage points that also have large residuals are *influential* — they could be distorting results. Look for points outside the Cook's distance dashed lines. Here no points are outside →

Written report for Q9:

A one-way analysis of variance indicated that the end-aisle display design has a significant effect on the percentage increase in sales, $F(2, 12) = 35.77$, $MSE = 0.263$, $p < 0.001$, $r^2 = 0.856$. Tukey's HSD test indicated that Display 3 produced significantly higher sales increases than both Display 1 ($p < 0.001$) and Display 2 ($p = 0.001$), while Displays 1 and 2 were not significantly different from each other ($p = 0.358$). Model diagnostic plots confirmed that all ANOVA assumptions were satisfied.

Annotation space:

3.9 Q10 — Brick Firing Temperature

3.9.1 Question

Four firing temperatures (100°C, 125°C, 150°C, 175°C) were tested to see if they affect brick density. Run a full analysis including handling an influential outlier.

3.9.2 Setup

```
library(outliers)

bricks <- read_tsv("brick.txt")
bricks$Temperature <- as.factor(bricks$Temperature)
glimpse(bricks)
```

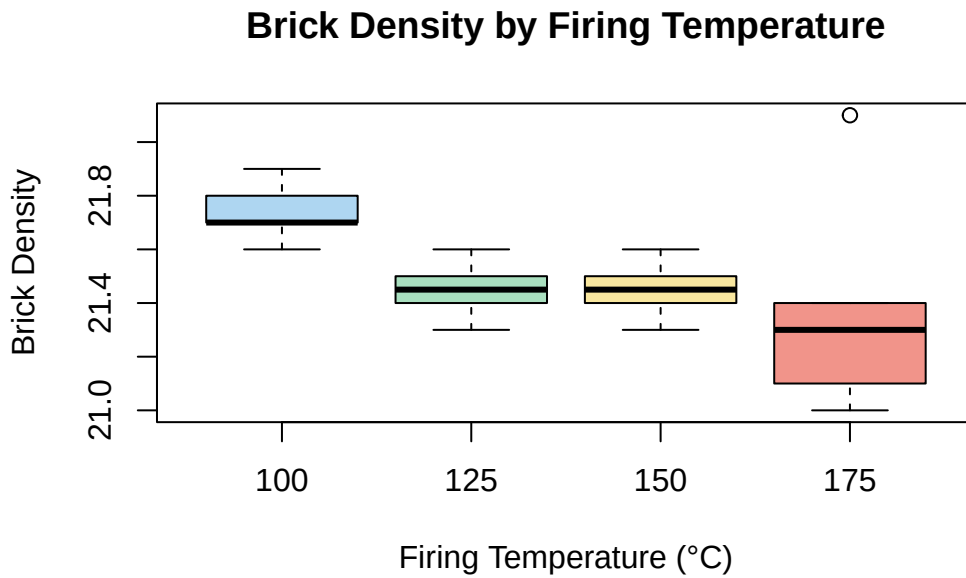
```
Rows: 40
Columns: 2
$ Temperature <fct> 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 100, 125, 125~
$ Density      <dbl> 21.8, 21.9, 21.7, 21.6, 21.7, 21.8, 21.9, 21.7, 21.6, 21.7~
```

3.9.3 Q10(i) — Exploratory Data Analysis

```
# Summary statistics
bricks %>%
  group_by(Temperature) %>%
  summarise(
    mean_Density = mean(Density),
    sd_Density   = sd(Density),
    var_Density  = var(Density),
    n            = n()
  )
```

```
# A tibble: 4 x 5
  Temperature mean_Density sd_Density var_Density     n
  <fct>         <dbl>      <dbl>      <dbl> <int>
1 100           21.7       0.107       0.0116    10
2 125           21.4       0.108       0.0117    10
3 150           21.4       0.108       0.0117    10
4 175           21.3       0.312       0.0973    10
```

```
# Boxplot
boxplot(bricks$Density ~ bricks$Temperature,
        xlab = "Firing Temperature (°C)",
        ylab = "Brick Density",
        main = "Brick Density by Firing Temperature",
        col = c("#AED6F1", "#A9DFBF", "#F9E79F", "#F1948A"))
```



Checking for outliers — Grubbs Test:

```
# Check overall dataset  
grubbs.test(bricks$Density)
```

Grubbs test for one outlier

```
data: bricks$Density  
G = 2.60652, U = 0.82133, p-value = 0.1317  
alternative hypothesis: highest value 22.1 is an outlier
```

```
# Check within the 175°C group specifically  
grubbs.test(bricks$Density[bricks$Temperature == "175"])
```

Grubbs test for one outlier

```
data: bricks$Density[bricks$Temperature == "175"]  
G = 2.50014, U = 0.22831, p-value = 0.004112  
alternative hypothesis: highest value 22.1 is an outlier
```

Grubbs test interpretation: - Overall dataset: outlier is **not** significant ($p > 0.05$) — no individual value is unusual enough to be an outlier when looking at all temperatures together.
 - Within the 175°C group only: the outlier **is** significant ($p < 0.05$) — observation 37 (density = 22.1) stands out as unusual within that temperature group.

Important: The observation 22.1 at 175°C looks suspicious. It may be a data entry error or a genuine anomaly. Since we don't have additional information, we need to run the analysis **both with and without** this point.

Checking equal variances:

```
fligner.test(bricks$Density ~ bricks$Temperature)
```

Fligner-Killeen test of homogeneity of variances

data: bricks\$Density by bricks\$Temperature

Fligner-Killeen:med chi-squared = 4.9654, df = 3, p-value = 0.1743

3.9.4 Q10(ii) — ANOVA (Full Dataset)

```
model_bricks <- aov(bricks$Density ~ bricks$Temperature)
summary(model_bricks)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
bricks\$Temperature	3	0.946	0.31533	9.539	8.97e-05 ***
Residuals	36	1.190	0.03306		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
summary.lm(model_bricks)
```

Call:

```
aov(formula = bricks$Density ~ bricks$Temperature)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.3200	-0.0725	-0.0300	0.0650	0.7800

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	21.74000	0.05749	378.127	< 2e-16 ***
bricks\$Temperature125	-0.29000	0.08131	-3.567	0.00104 **
bricks\$Temperature150	-0.29000	0.08131	-3.567	0.00104 **
bricks\$Temperature175	-0.42000	0.08131	-5.165	9.05e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1818 on 36 degrees of freedom

Multiple R-squared: 0.4429, Adjusted R-squared: 0.3965

F-statistic: 9.539 on 3 and 36 DF, p-value: 8.972e-05

3.9.5 Q10(iii) — Pairwise Comparisons (Full Dataset)

```
TukeyHSD(model_bricks)
```

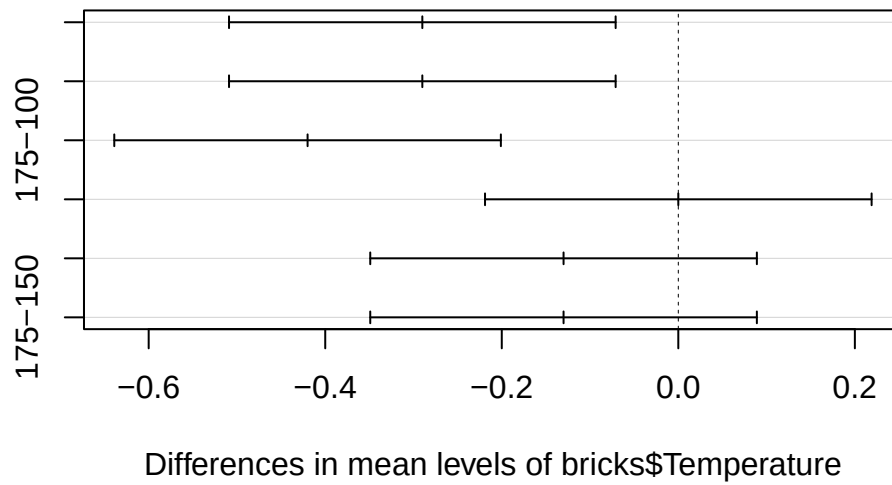
Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = bricks\$Density ~ bricks\$Temperature)

```
$`bricks$Temperature`  
      diff      lwr      upr      p adj  
125-100 -0.29 -0.5089829 -0.07101709 0.0055147  
150-100 -0.29 -0.5089829 -0.07101709 0.0055147  
175-100 -0.42 -0.6389829 -0.20101709 0.0000519  
150-125  0.00 -0.2189829  0.21898291 1.0000000  
175-125 -0.13 -0.3489829  0.08898291 0.3920570  
175-150 -0.13 -0.3489829  0.08898291 0.3920570
```

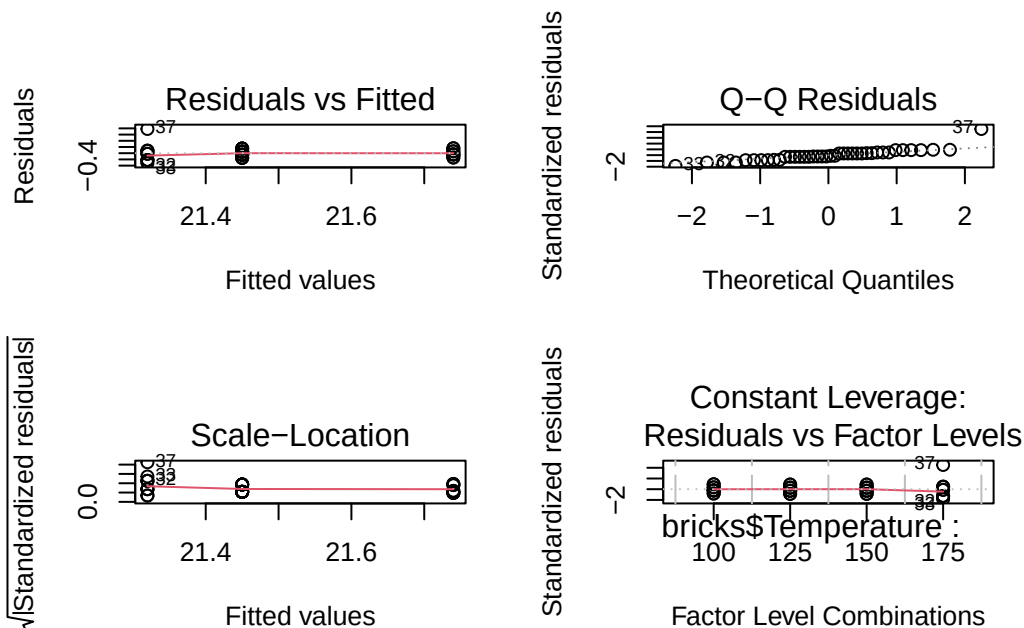
```
plot(TukeyHSD(model_bricks))
```

95% family-wise confidence level



3.9.6 Q10(vi) — Model Diagnostics (Full Dataset)

```
par(mfrow = c(2, 2))  
plot(model_bricks)
```



What to look for here: Observation 37 (the 22.1 density at 175°C) will likely show up as a point of high influence. Check the “Residuals vs Leverage” plot — if it falls outside the Cook’s distance lines, it is highly influential and the analysis should be rerun without it.

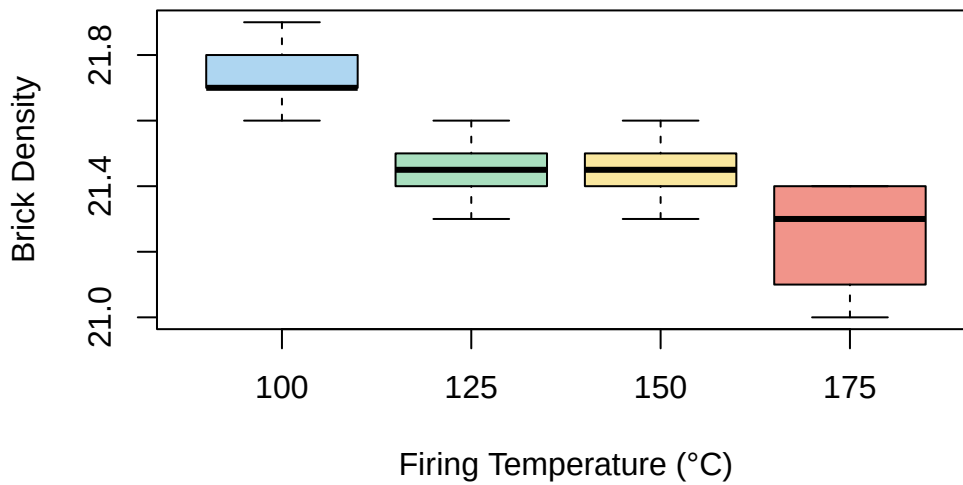
The diagnostic plots reveal that **observation 37 is highly influential**. We should remove it and rerun.

3.9.7 Reanalysis — Outlier Removed

```
bricks_rm <- bricks[-37, ]

# Check boxplots again
boxplot(bricks_rm$Density ~ bricks_rm$Temperature,
        xlab = "Firing Temperature (°C)",
        ylab = "Brick Density",
        main = "Brick Density by Temperature (Outlier Removed)",
        col = c("#AED6F1", "#A9DFBF", "#F9E79F", "#F1948A"))
```


Brick Density by Temperature (Outlier Removed)



```
# Test variance equality again  
fligner.test(bricks_rm$Density ~ bricks_rm$Temperature)
```

Fligner-Killeen test of homogeneity of variances

```
data: bricks_rm$Density by bricks_rm$Temperature  
Fligner-Killeen:med chi-squared = 3.1345, df = 3, p-value = 0.3713
```

```
# Refit model  
model_bricks_rm <- aov(bricks_rm$Density ~ bricks_rm$Temperature)  
summary(model_bricks_rm)
```

```
              Df Sum Sq Mean Sq F value    Pr(>F)        
bricks_rm$Temperature  3  1.240   0.4135   28.15 1.92e-09 ***  
Residuals              35   0.514   0.0147                  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary.lm(model_bricks_rm)
```

```
Call:
aov(formula = bricks_rm$Density ~ bricks_rm$Temperature)

Residuals:
      Min       1Q   Median       3Q      Max
-0.23333 -0.09167 -0.04000  0.06667  0.16667

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      21.74000    0.03832  567.299 < 2e-16 ***
bricks_rm$Temperature125 -0.29000    0.05420   -5.351 5.54e-06 ***
bricks_rm$Temperature150 -0.29000    0.05420   -5.351 5.54e-06 ***
bricks_rm$Temperature175 -0.50667    0.05568   -9.100 9.43e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1212 on 35 degrees of freedom
Multiple R-squared:  0.707, Adjusted R-squared:  0.6819
F-statistic: 28.15 on 3 and 35 DF,  p-value: 1.916e-09
```

```
# Tukey again
TukeyHSD(model_bricks_rm)
```

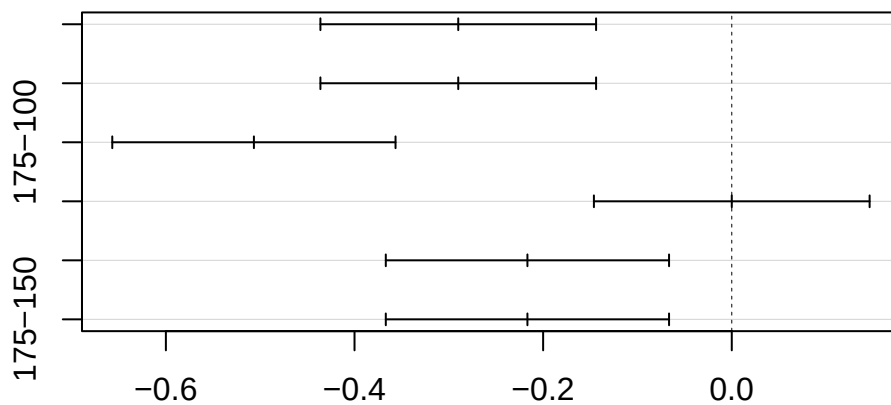
```
Tukey multiple comparisons of means
 95% family-wise confidence level
```

```
Fit: aov(formula = bricks_rm$Density ~ bricks_rm$Temperature)
```

```
$`bricks_rm$Temperature`
      diff      lwr      upr      p adj
125-100 -0.2900000 -0.4361598 -0.14384024 0.0000318
150-100 -0.2900000 -0.4361598 -0.14384024 0.0000318
175-100 -0.5066667 -0.6568315 -0.35650179 0.0000000
150-125  0.0000000 -0.1461598  0.14615976 1.0000000
175-125 -0.2166667 -0.3668315 -0.06650179 0.0023099
175-150 -0.2166667 -0.3668315 -0.06650179 0.0023099
```

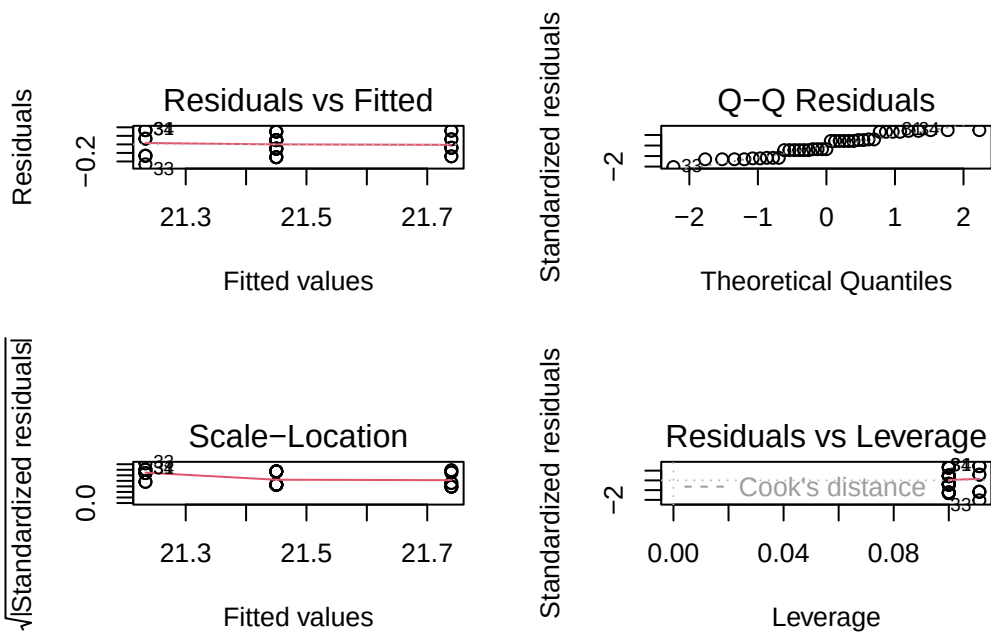
```
plot(TukeyHSD(model_bricks_rm))
```

95% family-wise confidence level



Differences in mean levels of bricks_rm\$Temperature

```
# Diagnostics
par(mfrow = c(2, 2))
plot(model_bricks_rm)
```



3.9.8 How to Write Up Both Results

Since we have no additional information about the outlier (observation 37, density = 22.1 at 175°C), and it is a plausible value for brick density, we report both analyses:

With outlier: A one-way ANOVA [report F, df, MSE, p, η^2]. Tukey's HSD indicated [report which groups differ].

Without outlier: Removing observation 37 changes the pairwise comparisons [describe how results change]. Both sets of results are presented as the outlier cannot be definitively excluded.

Exam tip: When an outlier is influential, always (a) report it, (b) investigate it, (c) run the analysis both ways, and (d) explain what changes.

Annotation space:

4 Summary Points — Key Takeaways

- **Factor** = what you control. **Level** = the settings. **Treatment** = one specific combination.
 - **Randomized block design** controls for a known nuisance variable by including it as a “block.”
 - **Multiple testing** inflates Type I error: $P(\text{at least one error}) = 1 - (1 - \alpha)^k$.
 - **ANOVA assumptions:** independence, equal variances (tested with Fligner or Levene), approximately normal residuals.
 - **ANOVA hypotheses:** H_0 : all group means equal. H_a : at least one differs.
 - **F statistic** = $MS_{\text{Treatment}} / MS_{\text{Residuals}}$. Big F $\rightarrow$ more evidence against H_0 .
 - **Degrees of freedom:** Treatment = $k - 1$. Residuals = $N - k$.
 - $\eta^2 = SS_{\text{Treatment}} / SS_{\text{Total}}$. Proportion of variation explained. Exam standard: report alongside F, df, MSE, p.
 - **Tukey HSD:** controls family-wise error when doing all pairwise comparisons. Groups sharing a superscript letter are not significantly different.
 - **The ANOVA model:** $\hat{y} = \mu + \alpha_1 x_1 + \alpha_2 x_2 + \dots$ where μ = mean of reference group and α coefficients = differences from reference group.
 - **Diagnostics:** Check Residuals vs Fitted (patterns?), Q-Q plot (normality?), Scale-Location (constant variance?), Leverage plot (influential points?).
 - **Influential outliers:** Always run the analysis with AND without them if they appear influential. Report both and explain.
-

5 Quick Formula Reference

```
#  $\eta^2$  (Effect Size)
eta_sq <- SS_Treatment / (SS_Treatment + SS_Residuals)

# Degrees of freedom
df_treatment <- k - 1      # k = number of groups
df_residuals <- N - k      # N = total observations

# Mean Squares
MS_Treatment <- SS_Treatment / df_treatment
MS_Residuals <- SS_Residuals / df_residuals

# F statistic
F_stat <- MS_Treatment / MS_Residuals

# Type I error (multiple tests)
P_at_least_one_error <- 1 - (1 - alpha)^num_tests
```

General annotations — add your own notes, mnemonics, or diagrams below: ““